

ENV 4120

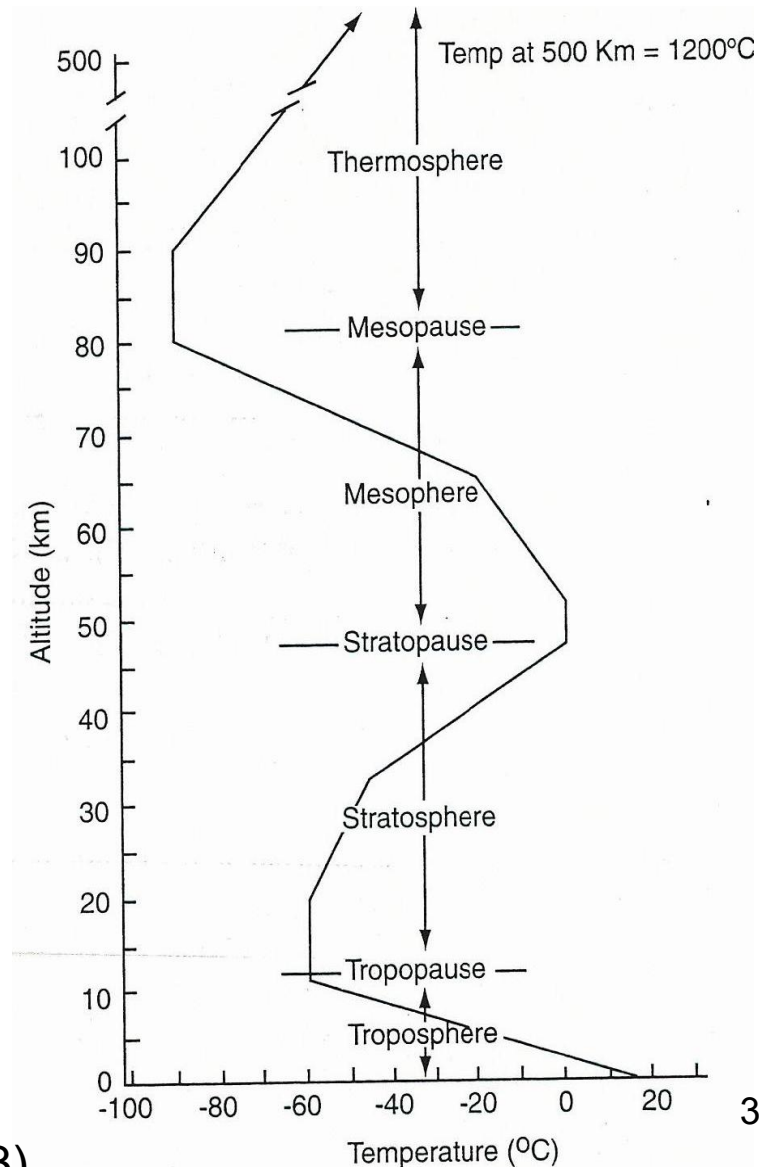
Air Pollution and Hazardous Waste Control

Today's Topics

- Basic meteorology related to air quality
- Expectation
 - Refresh knowledge on atmosphere motion at different scales and wind rose map
 - Understand the structure (density, pressure, temperature) of atmosphere, lapse rate (dry, wet and environmental), inversion, stability class
 - Be able to convert temperature and pressure among different units, calculate wind speed at given height

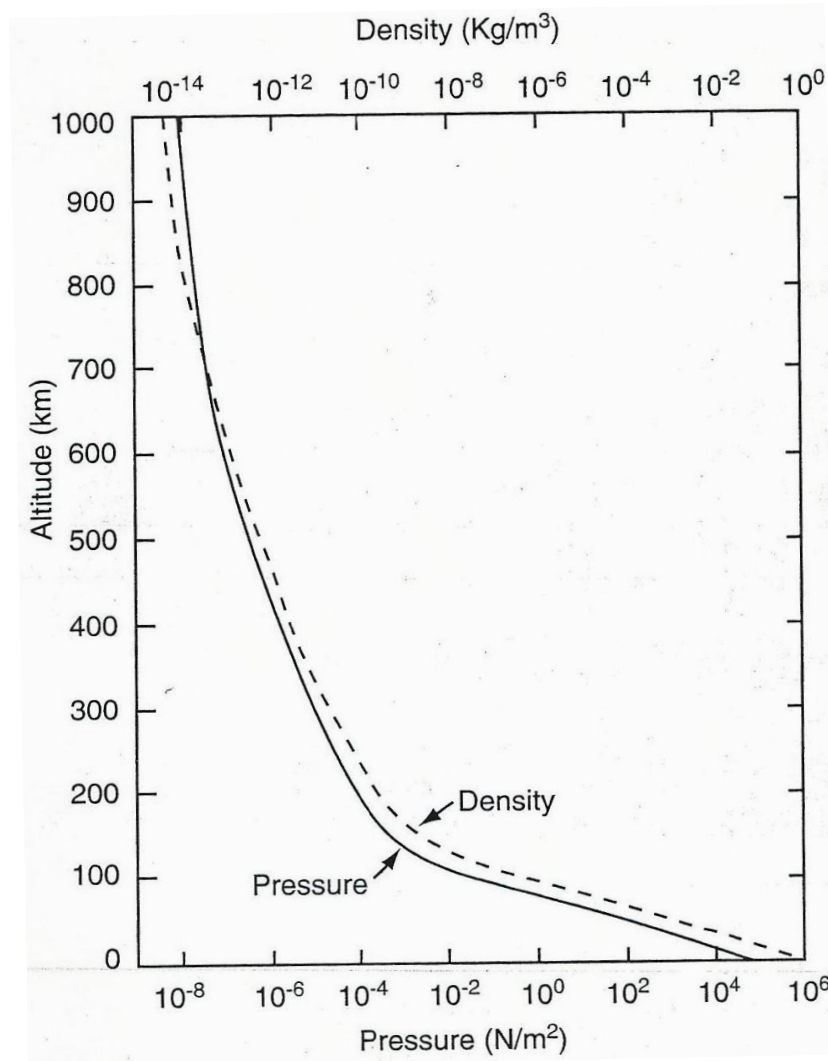
Atmospheric Temperature Structure

- Troposphere
 - Height ↑ Temp ↓
 - Rapid vertical mixing
- Stratosphere
 - Height ↑ Temp ↑
 - Inversion
 - Stable



(Godish, 2003)

Atmospheric Density and Pressure



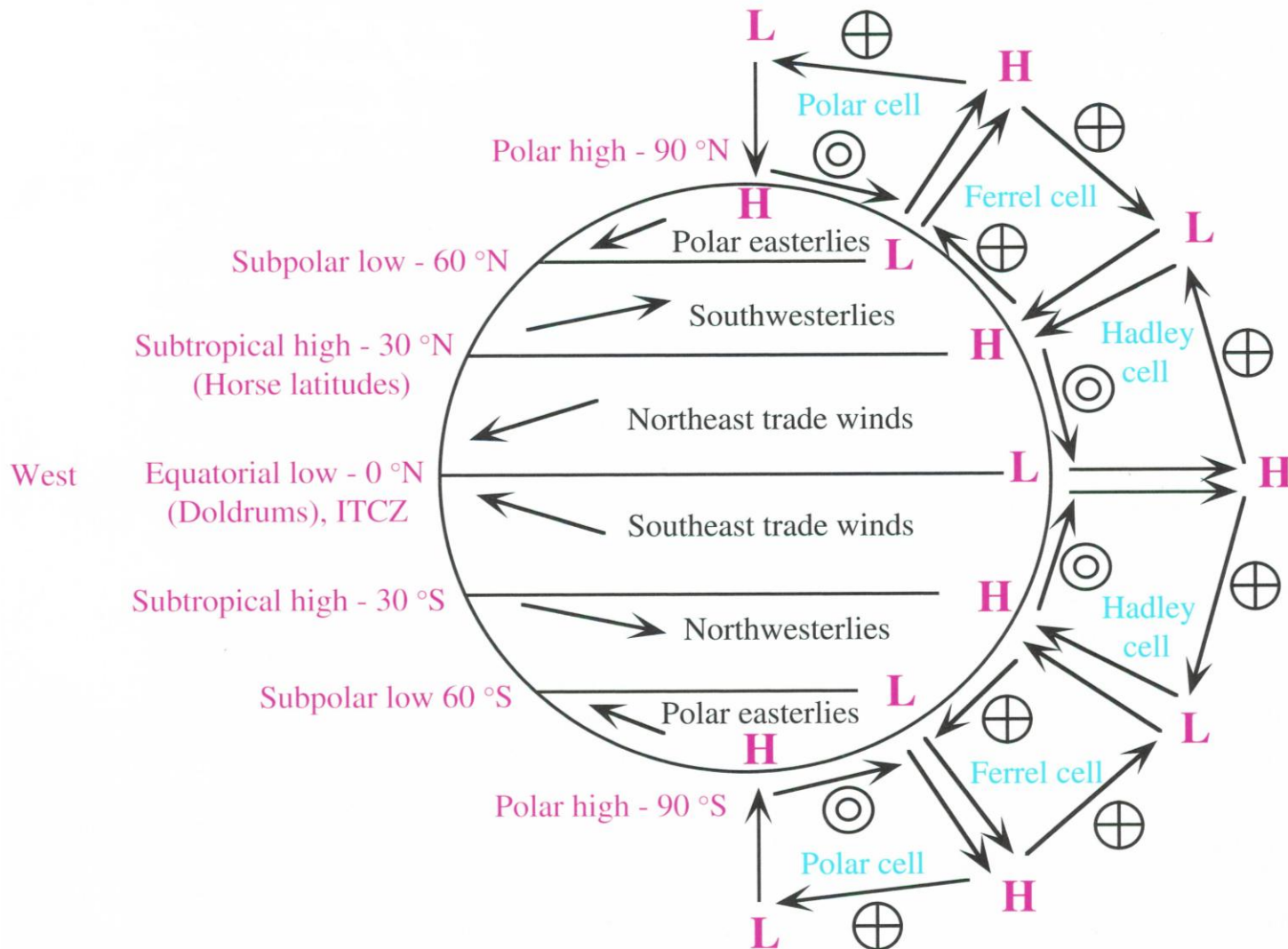
Conversion of temperature & pressure

- Temperature: Kelvin (K), Celsius ($^{\circ}\text{C}$), Fahrenheit ($^{\circ}\text{F}$)
 - $\text{K} = 273.15 + ^{\circ}\text{C}$
 - $^{\circ}\text{F} = 1.8^{\circ}\text{C} + 32$
 - $^{\circ}\text{C} = (^{\circ}\text{F} - 32)/1.8$
- Pressure
 - $1 \text{ atm} = 1013.25 \text{ mbars} = 1.01325 \times 10^5 \text{ Pa}$
 $= 14.69 \text{ lb/in}^2 \text{ (psi)} = 760 \text{ mmHg}$
 $= 10332 \text{ mmH}_2\text{O}$

Pollution transport

- After emitted/formed, pollutants need to be transported to get in contact with “receptor”
 - Human, certain part of eco-system etc..
- Pollution transport depends on meteorology
- Pollutants can be removed or formed during the process
- The distance of transport differ drastically depends on pollutant characteristics

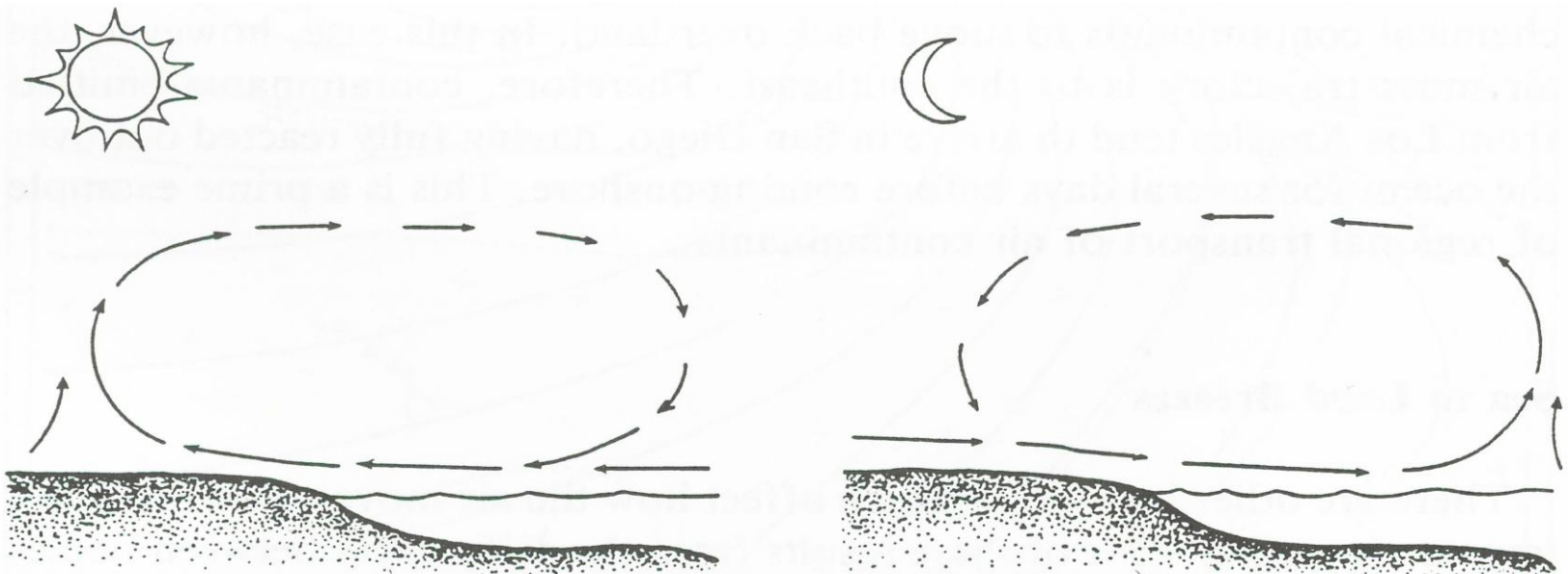
Large Scale Motion: General Circulation pattern



(Jacobson, 2002)

Smaller scale motion

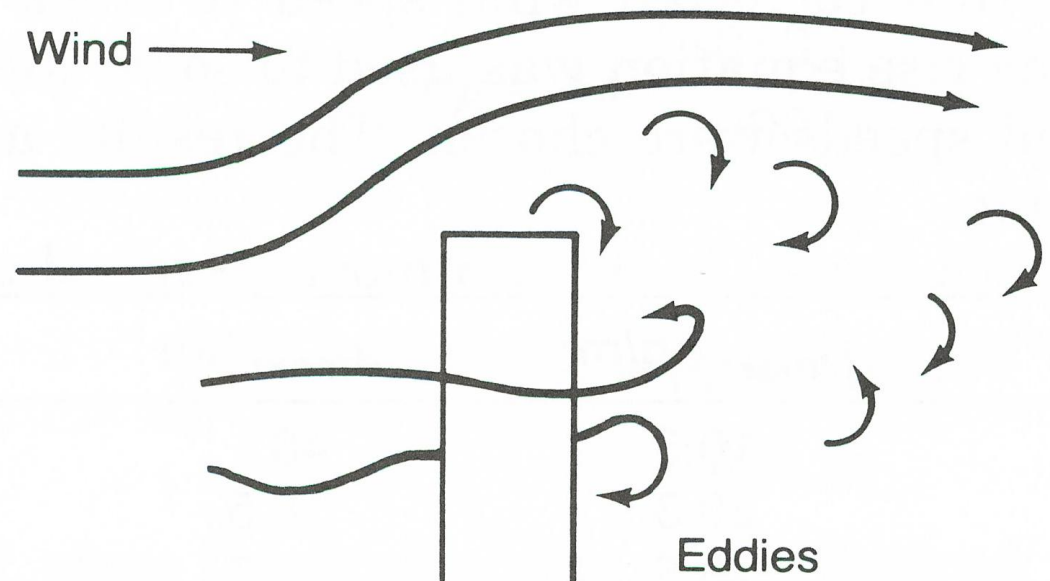
- Impact by variable heating and topography
 - valley and mountain breezes
 - sea and land breezes
 - urban heat island flow



Example of land sea breeze (Griffin, 1994)

Even smaller scale

- Turbulence
 - Random / irregular fluctuations on top of mean flow
 - Circular eddies at smaller scale than mean flow
- Important to
 - mixing
 - dilution
- Pollution hot spot



Gone with the wind – Wind Rose

Horizontal mean flow

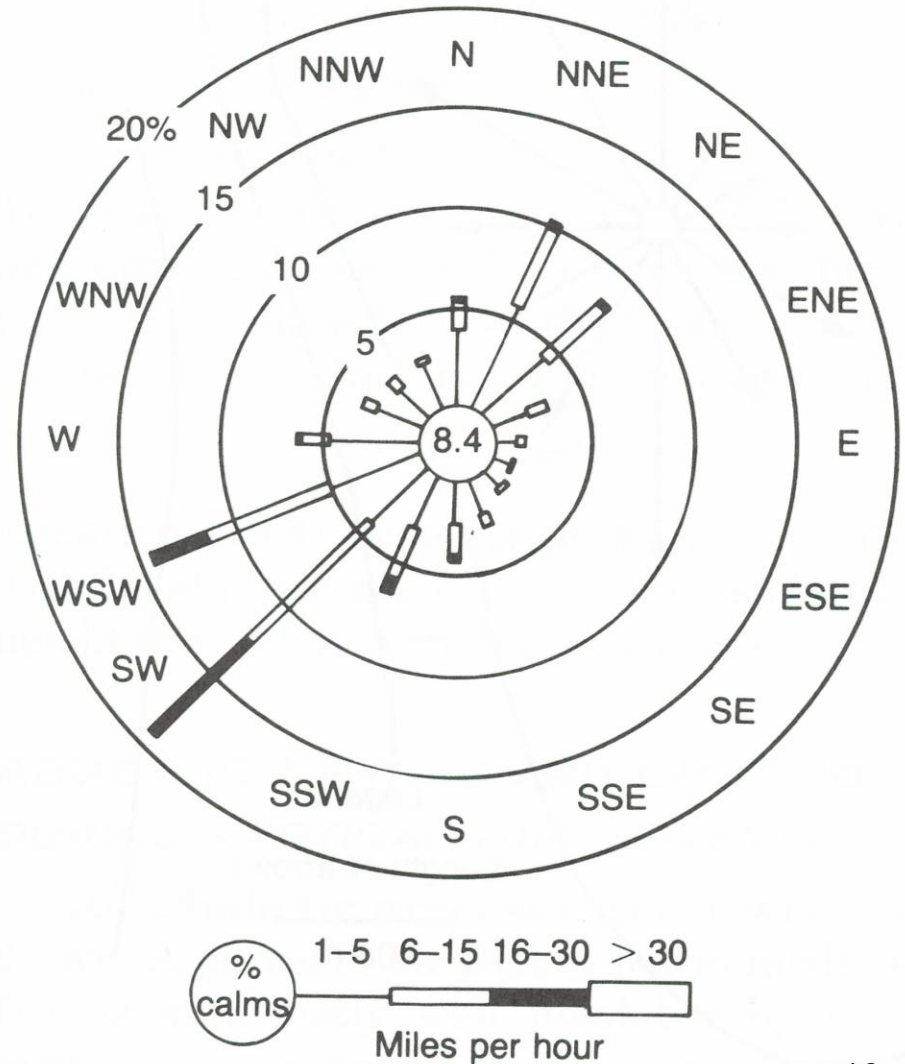
Direction

Where wind is coming from
(angle in degrees clockwise
from the north)

Direction of pollutant transport

Speed

transports pollutants further
dilutes emissions
creates turbulent mixing
bends plumes



Understanding Stability:

Vertical air motion

- Buoyancy
 - Air hotter than surrounding is less dense (and rises)
- Rising and falling air leads to
 - vertical overturning and mixing
 - pollutant dispersal/dilution due to turbulence
- Atmosphere can enhance or suppress this
 - Stable – atmosphere resists vertical motion
 - Unstable – atmosphere enhances vertical motion
 - Neutral – atmosphere does neither

Understanding Stability:

Vertical air motion

- Say an air parcel rise from the ground....
 - Stable → atmosphere hamper the movement
 - Unstable → atmosphere facilitate the movement
 - Neutral → atmosphere does neither
- The environmental temperature structure is important

Lapse rate

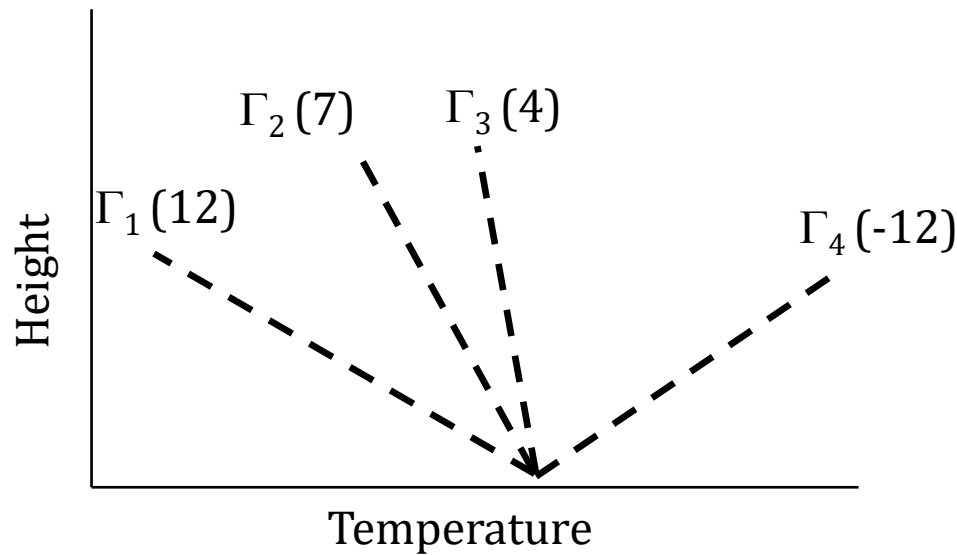
- The change in temperature with height

$$\Gamma = -\frac{\Delta T}{\Delta z}$$

Γ : lapse rate (K/km or °C/km).

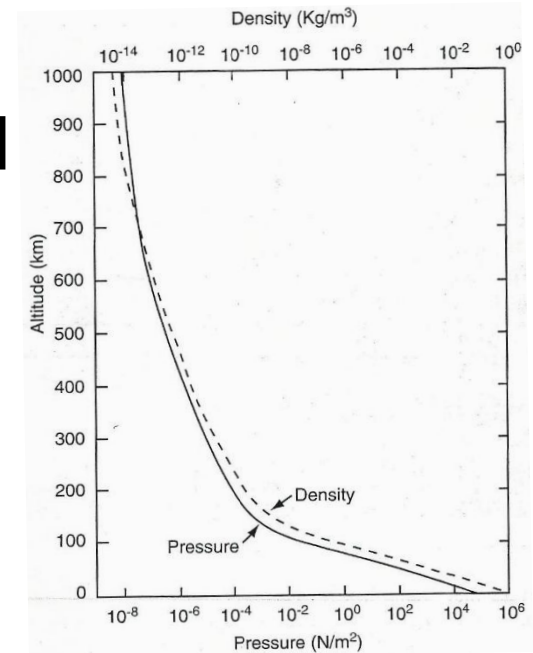
ΔT : temperature changes (K or °C) over height Δz (km)

negative if decreases with altitude increasing.



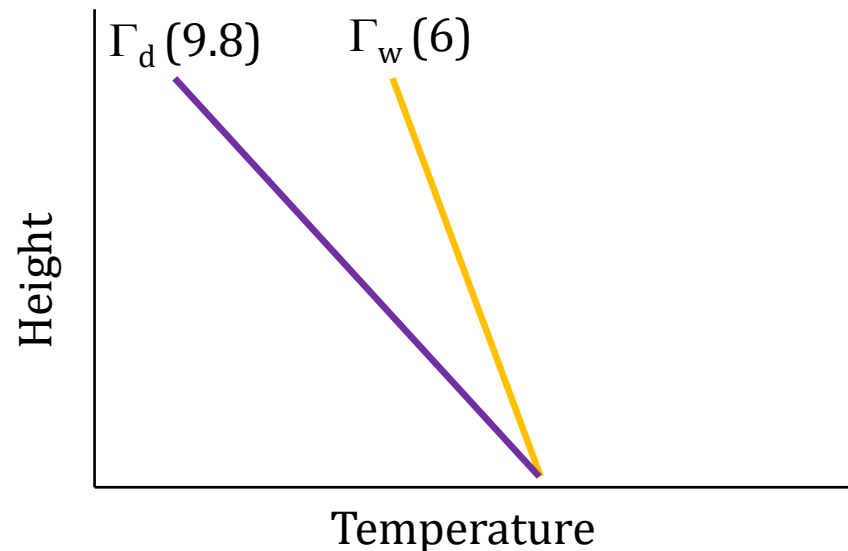
Rise of air parcel

- Suppose we have a parcel of air ascending from ground, *adiabatically*.....
 - Adiabatic Process: Process by which no energy is exchanged between a system (parcel of air) and its surroundings (atmosphere).
- Height $\uparrow$ pressure $\downarrow$ parcel expand
Air parcel $T \downarrow$
 - Due to energy balance
 - *No heat exchange between parcel & environment (adiabatic)*



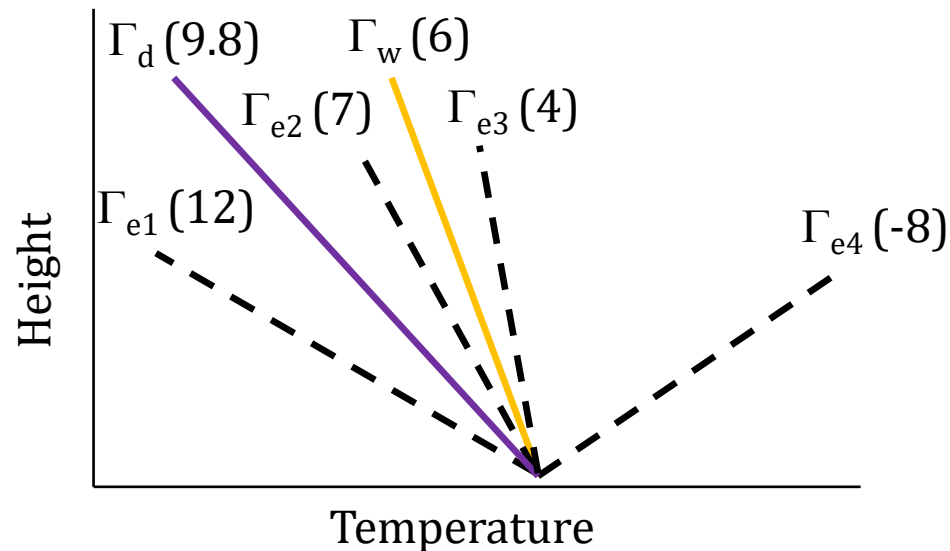
Rise of air parcel

- Dry air parcel: lose $9.8\text{ }^{\circ}\text{C}/\text{km} = \Gamma_d$
- Wet air parcel (saturated with moist) lose less temperature than dry air parcel
 - $T \downarrow$, moist $\rightarrow$ water $\rightarrow$ release heat
 - Γ_w varies $\approx 6\text{ }^{\circ}\text{C}/\text{km}$ in this class



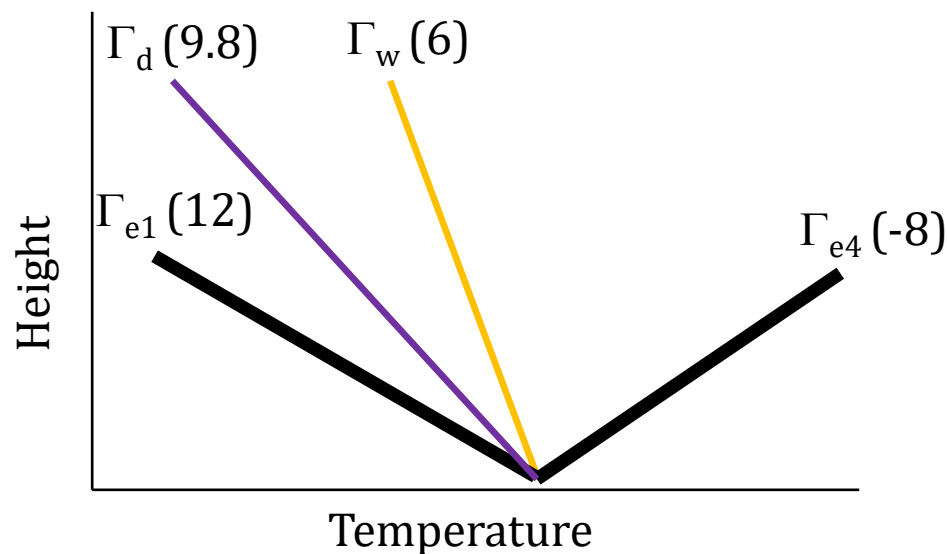
Rise of air parcel

- Lapse rate of lower troposphere
 - Environment lapse rate (Γ_e), varies greatly
- Air warmer than surrounding gain buoyancy
- How different Γ_e impact air parcel rising (vertical movement)?



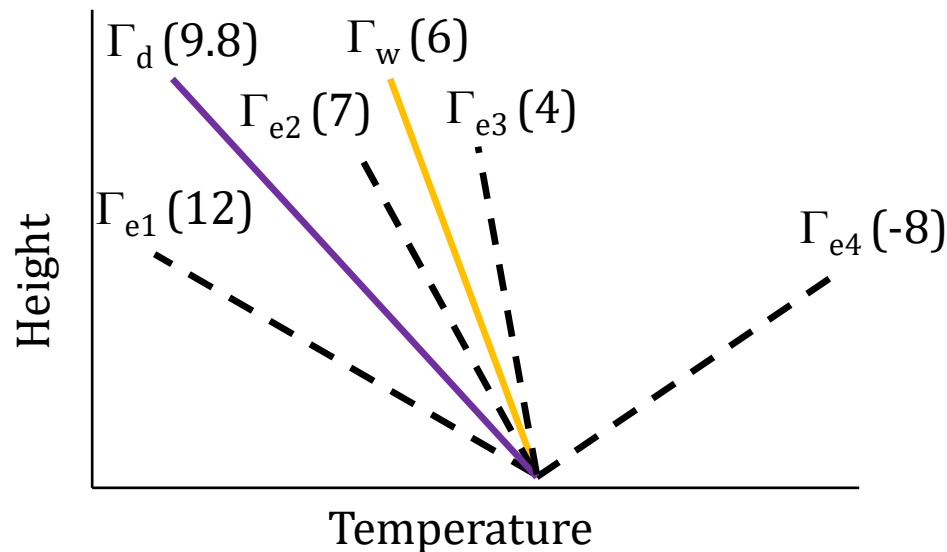
Rise of air parcel

- $\Gamma_e > \Gamma_d$: Absolutely unstable (Γ_{e1})
 - T of air parcel always > T of surrounding → keeps rising
- $\Gamma_e < \Gamma_w$: Absolutely stable (Γ_{e4})
 - T of air parcel always < T of surrounding → stop rising



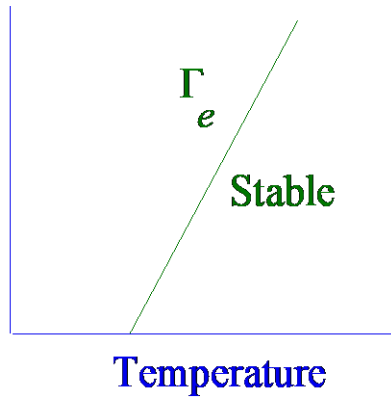
Rise of the air parcel

- ☐ $\Gamma_e > \Gamma_d$: Absolutely unstable (Γ_{e1})
- ☐ $\Gamma_e = \Gamma_d$: Dry neutral
- ☐ $\Gamma_d > \Gamma_e > \Gamma_w$: Conditionally unstable (Γ_{e2})
- ☐ $\Gamma_e = \Gamma_w$: Wet neutral
- ☐ $\Gamma_e < \Gamma_w$: Absolutely stable (Γ_{e3}, Γ_{e4})

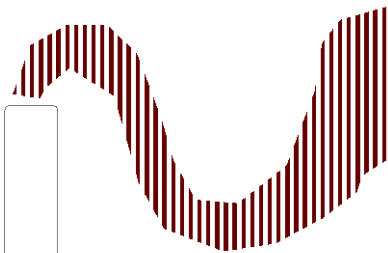
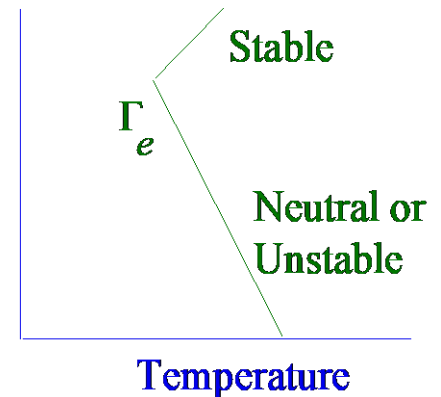
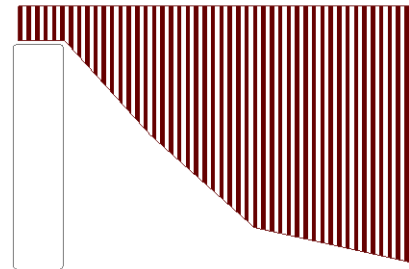


Effects on Pollutant Releases

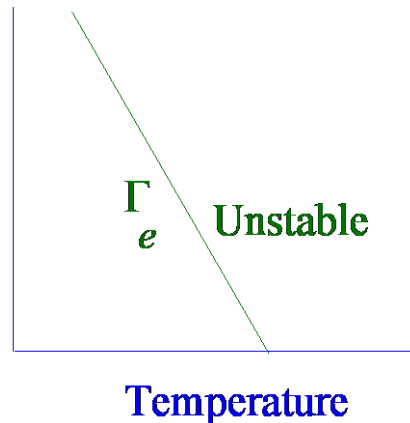
Fanning



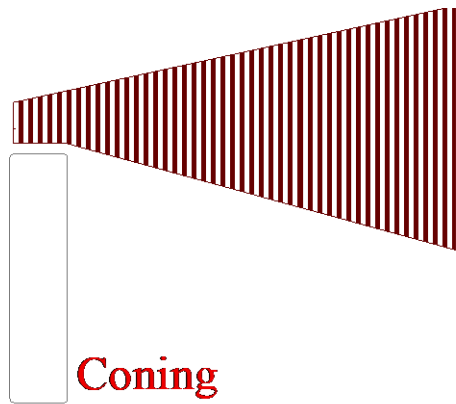
Fumigating



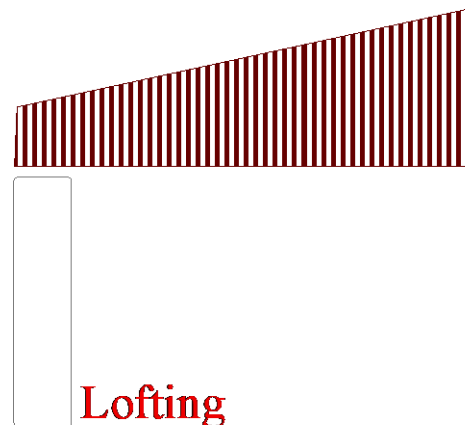
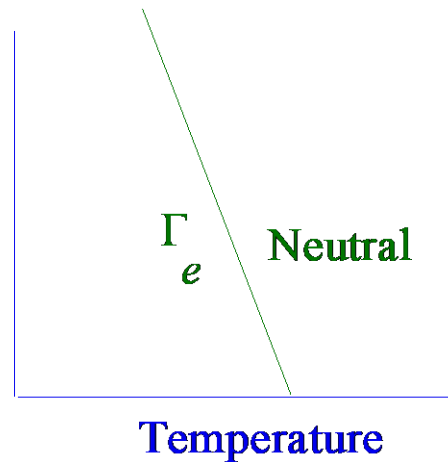
Looping



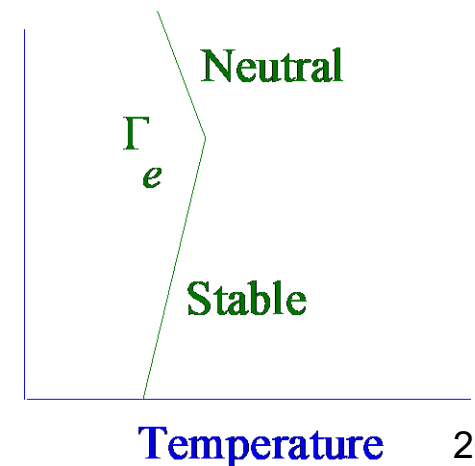
Effects on Pollutant Releases



Coning

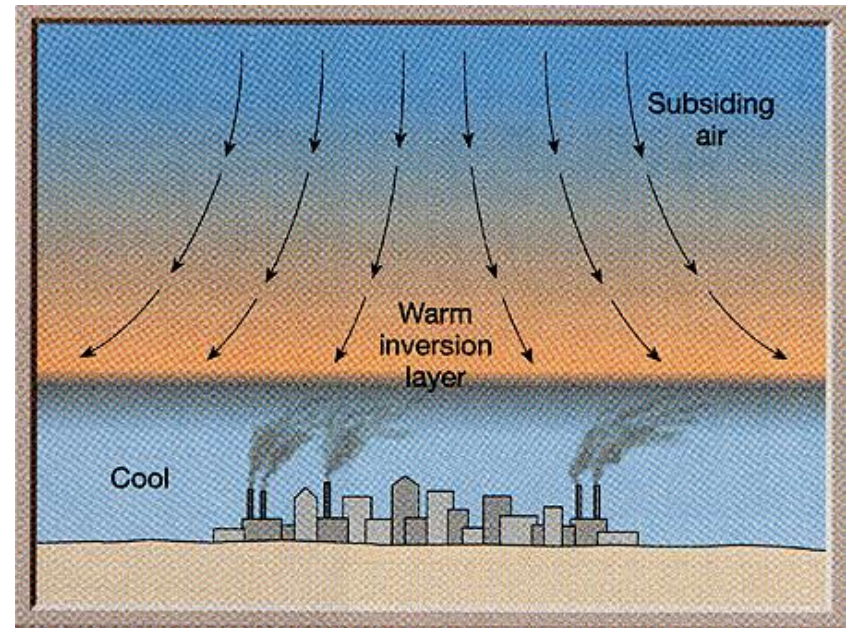


Lofting



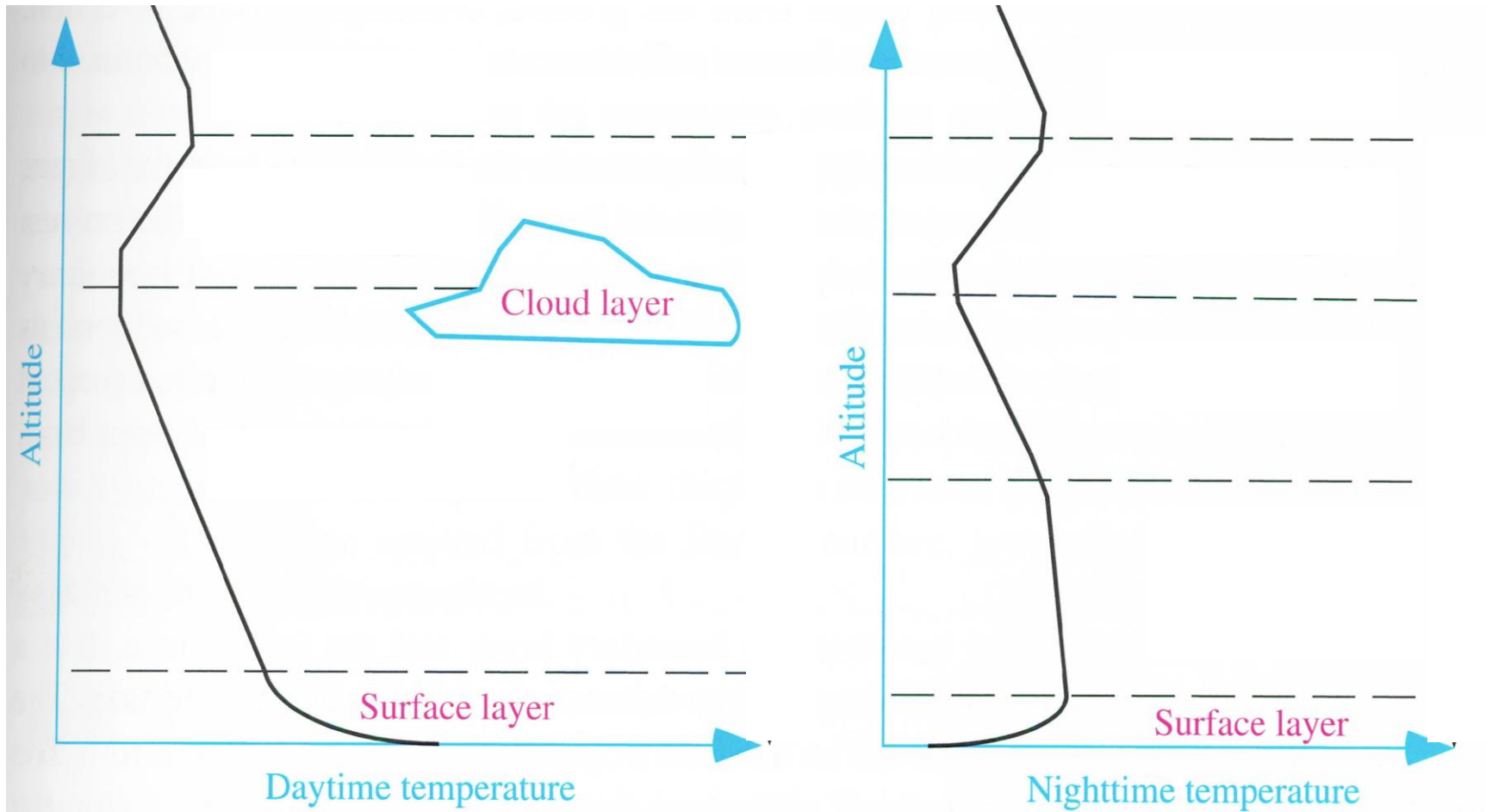
Inversions

- Definition: a negative environmental lapse rate
 - Temperature increases with height
- Always stable
- Why important?
 - Can trap air and pollutants
- Types and effects
 - Frontal
 - Nocturnal (radiation)
 - Subsidence



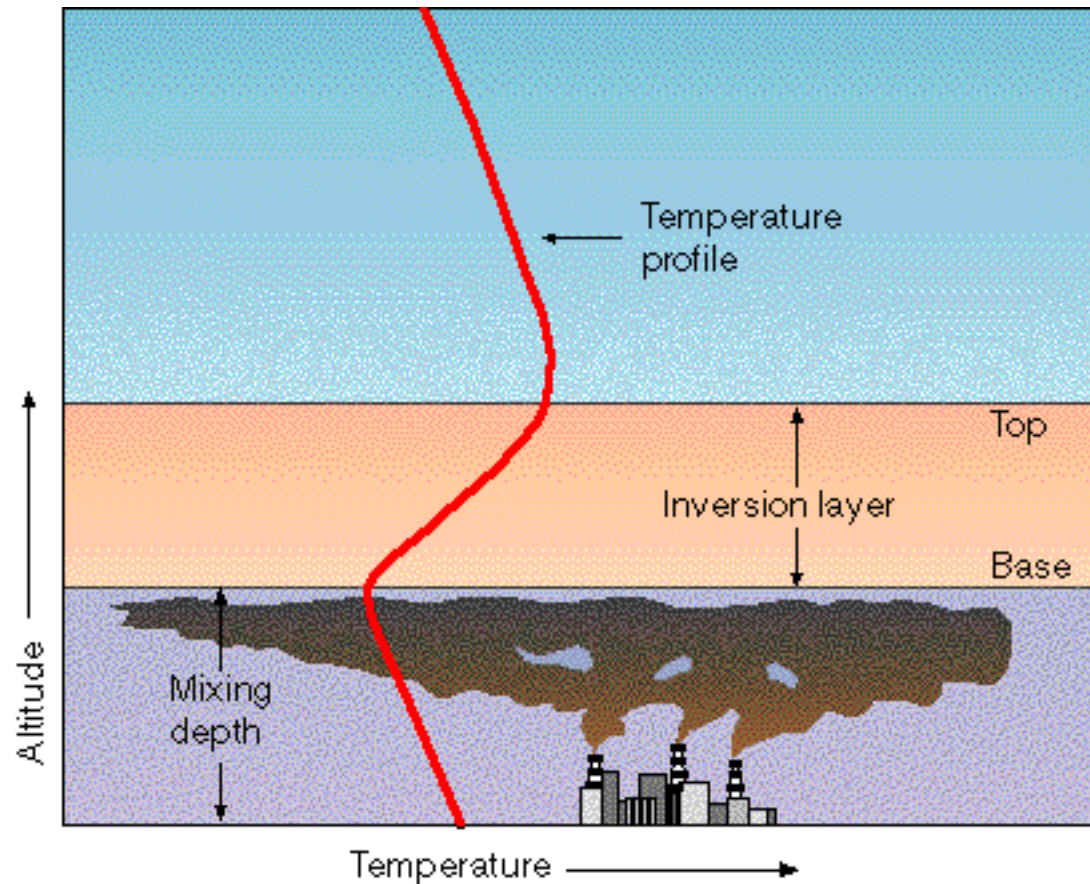
Tropospheric Temperature Structure

Diurnal Variation



Mixing Depth /Height

- Height to which air is vigorously mixed
- Definable if unstable or neutral below and stable above



Categorizing Stability: Classes

day: more stable as u increases
night: less stable as u increases

more stable with less clouds

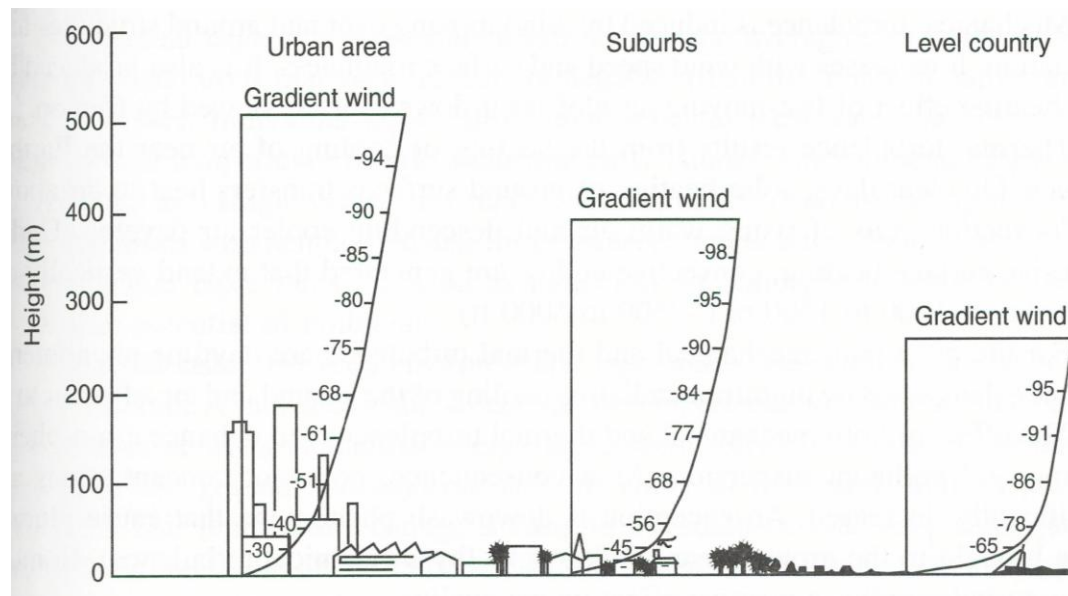
Surface Wind				Night-Time Cloudiness	
(measured at 10 m)	Day-Time Insolation			Thinly Overcast or $\geq$ 4/8 Cloudiness	
m s^{-1}	Strong	Moderate	Slight		$\leq 3/8$ Cloudiness
<1	A	A-B	B	-	-
2-3	A-B	B	C	E	F
3-5	B	B-C	C	D	E
5-6	C	C-D	D	D	D
>6	C	D	D	D	D

A-extremely unstable, B-moderately unstable, C-slightly unstable, D-neutral, E-slightly stable, F-moderately stable

(Hanna et al., 1982, Handbook on Atmospheric Diffusion. U.S. Dept. of Energy).

Wind: Vertical

- Wind direction changes vertically
- Wind speed also changes
 - Lower speed near ground due to friction
 - Different land use → different friction



Example vertical wind speed profile (Godish, 2003)

Wind speed profile power law

- Wind speed vs altitude (power law)

$$\frac{u}{u_0} = \left(\frac{z}{z_0} \right)^p$$

u : wind speed (m/s) at height z (m)

u_0 : reference wind speed, typically measured at 10 m (z_0)

p : coefficient, varies with surface roughness and stability
(0.15-0.55) (We'll learn about stability later)

Look up table for p value

Stability Class	Urban	Rural
A	0.15	0.07
B	0.15	0.07
C	0.2	0.1
D	0.25	0.15
E	0.3	0.35
F	0.3	0.55